Luma-1 prototype parts specification (Macro update)

09 November 2026 | Revision B | Request for Quotation & Manufacturing Drawing Review

Design Basis: Slow level-ground humanoid prototype walking, 30.0 kg all-up mass, 48 V electrical architecture. Dimensions established by 105-candidate simulation optimization and 56 adverse validation runs.

1. Carbon-Fiber Leg Rods (8 pieces total per robot)

Component	Qty	Cut length	Outer diameter	Joint spacing	Target tolerance
Thigh carbon rod	4	204.00 mm	20.0 mm nominal	164.0 mm	±0.25 mm (pair matched ±0.10 mm)
Shin carbon rod	4	161.00 mm	20.0 mm nominal	247.0 mm	±0.25 mm (pair matched ±0.10 mm)

- **Dimensions:** Thigh 204.0 mm, Shin 161.0 mm cut lengths. Outer diameter 20.0 mm nominal to fit existing chassis clamps.
- **Tolerances:** Length tolerance ±0.25 mm. Matched pair length difference ≤ 0.10 mm.
- **Structural Capacity:** Working axial capacity ≥ 3.0 kN per finished rod assembly in tension and compression.

2. Passive Ankle Coilovers (4 units total per robot: 2 Front, 2 Rear)

Parameter	Front shock requirement	Rear shock requirement
Specification parameter	Front shock absorber (qty 2)	Rear shock absorber (qty 2)
Eye-to-eye center distance: fully extended	99.8 mm	119.9 mm
Eye-to-eye center distance: fully compressed	67.8 mm	87.9 mm
Usable mechanical stroke	32.0 mm (unobstructed by bump stop)	32.0 mm (unobstructed by bump stop)
Spring rate target	28.0 N/mm (159.9 lb/in) ±10%	29.5 N/mm (168.4 lb/in) ±10%
Damping coefficient target (linear)	260 N·s/m ±20%	510 N·s/m ±20%
Preload setting at full extension	0.5 mm (14.0 N initial force)	0.5 mm (14.8 N initial force)
CAD standing pose spring load	462.0 N	487.0 N
Required preload adjustment range	0.25 to 1.50 mm	0.25 to 1.50 mm
Required damping adjustment range	±20% via external adjuster	±20% via external adjuster
Axial proof load target (compression/tension)	1,600 N (1.6 kN)	1,900 N (1.9 kN)
Minimum ultimate structural load capacity	2,500 N (2.5 kN)	3,000 N (3.0 kN)

- **Usable Stroke:** 32.0 mm mechanical motion before hard stop (must not be consumed by bump stop).
- **Tolerances:** Eye-to-eye extended length ±0.25 mm. Spring rate ±10%. Damping force ±20% over 0.05 to 0.55 m/s.
- **Clearance:** No coil bind within full 32 mm mechanical stroke under maximum preload.

Supplier acceptance criteria & drawing checklist

Shock dynamic force envelope

Supplier must provide dynamometer force-velocity curves in compression and rebound across the velocity envelope below:

Shaft velocity	Front damping force (c = 260 N·s/m)	Rear damping force (c = 510 N·s/m)
0.05 m/s	13.0 N	25.5 N
0.10 m/s	26.0 N	51.0 N
0.30 m/s	78.0 N	153.0 N
0.52 m/s (peak observed)	135.2 N	265.2 N

Component structural acceptance loads

- **Front Shock Proof Load:** 1,600 N axial proof load (tested in compression and tension without permanent set).
- **Rear Shock Proof Load:** 1,900 N axial proof load (tested in compression and tension without permanent set).
- **Ultimate Load:** Minimum 2.5 kN (front) and 3.0 kN (rear) before rupture or disconnection.

Engineering drawings required to close RFQ

1. Detailed dimensional drawings showing mounting eye bore diameter, eye width, and spherical bearing articulation angle.
2. Spring free length, solid height, wire diameter, outer diameter, and clear envelope around shock body.
3. Elastomeric bump-stop dimensions and engagement travel within the 32 mm stroke.
4. Carbon tube wall thickness, fiber layup, insert bonding length, and clamp torque limits.

Procurement & Quotation Notice

This brief constitutes a formal prototype RFQ specification. Quantities for initial prototype batch: **8 carbon rods and 4 shock absorbers per robot** (recommend 2× sets for testing). Direct questions to Isla Technologies engineering.

All simulation verification records, CAD references, and test manifests are archived in the project engineering repository.